

Desk Number _____
Student Number _____
Student Name _____



School of Information & Physical Sciences PRACTICE EXAMINATION

Semester X, 20XX

SENG1050 Web Technologies

This paper is for CALLAGHAN students.

Examination Duration: 120 minutes

Reading Time: 10 minutes

This exam has NINETEEN (19) questions

Exam Conditions:

This is a FORMAL examination

This is a RESTRICTED OPEN book examination

Notes Permitted

Non Programmable Calculators Permitted

Materials Permitted In The Exam Venue:

1 A4 double sided sheet of handwritten or typed notes (Memory Aid sheet)

Non programmable calculators are permitted

Materials To Be Supplied To Students:

1 x 12 Page Answer Booklet

1 x GPAS

Special Instructions:

Answer all questions in **Section A**, **Section B** and **Section C**.

- Section A contains TEN (10) multiple-choice questions which are worth 2 marks each (worth 20 marks in total). Choose only one answer per question. Answer all the questions in this section by filling the GPAS multiple-choice answer sheet provided using a pencil. Also, make sure to encode your name and student number on the scan form using a pencil.
- Section B contains SIX (6) short answer questions (worth 50 marks in total). Note that each question may have multiple sub-questions. Answer all the questions in this section by filling the 12-page answer booklet provided.
- Section C contains THREE (3) long answer questions (worth 30 marks in total). Note that each question may have multiple sub-questions. Answer all the questions in this section by filling the 12-page answer booklet provided.

If you believe there is an error in the exam, please notify the exam invigilator. The exam invigilator can contact the course coordinator at adrian.tan10@newcastle.edu.au.

SECTION A MULTIPLE-CHOICE (20 marks, 2 marks per question)

Revise the weekly practice quizzes and review questions (questions presented at the start of lecture using Poll Everywhere).

Note that these revision questions may be more challenging and cover a wider scope.

SECTION B SHORT ANSWER (50 marks)

1. **(8 marks)**
 - a. **(4 marks)** Discuss the role of the Content Management Application (CMA) in a Content Management System (CMS).
 - b. **(4 marks)** Discuss the role of the Content Delivery Application (CDA) in a Content Management System (CMS).
2. **(6 marks)**
 - a. **(2 marks)** Compare the use of HTTP GET method and HTTP POST method during form submission.
 - b. **(2 marks)** Describe when the HTTP GET method is used.
 - c. **(2 marks)** Describe when the HTTP POST method is used.
3. **(8 marks)**
 - a. **(4 marks)** Identify and briefly describe **TWO (2)** specific principles of good web design for accessibility.
 - b. **(4 marks)** For each principle listed above, give a specific example of how it can be applied to the website you created for the web-based assignments in this course.

SECTION C LONG ANSWER (30 marks)

1. **(13 marks)**
 - a. **(3 marks)** Write a **valid HTML document** that will display two number inputs, and a button that calls a JavaScript function 'addition()' when the user clicks on it.
 - b. **(10 marks)** Write the JavaScript function 'addition()' for the HTML document you created in part a. This function would perform **addition** on the two input numbers. Give the output in a popup box using the alert() function.
2. **(8 marks)** A HTML form has a four numerical inputs, each with name attributes containing the values "num1", "num2", "num3", and "num4" respectively. The form is submitted via a HTTP GET request to a PHP script called 'submit.php'. Write the PHP script 'submit.php' that returns the **second-largest number** entered.
3. **(9 marks)** Use the Huffman coding method to encode the ASCII string: 'goodluck' (without the quotes), **showing all working**, and provide the compressed message and the compression amount as a percentage and as a ratio in exact form.

END OF EXAMINATION